INTRODUCTION TO R II - UMAINE

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The Jackson Laboratory

12.5 HOURS • 5 Modules • 1-credit hour
Completion of Intro to R I • Biostatistics I
Part of Foundational Data Science Training Series

Course Description

"How do we transition from writing basic code to building robust, reproducible pipelines that actually 'do science'"?

This course is the second half of the Introduction to R sequence. Building directly on R I, we shift our focus from R as a programming language to R as a data-analysis platform. We explore the tidyverse philosophy, master ggplot2 for publication-ready visualization, and work through the statistical toolkit used every day in biological research: hypothesis testing, goodness-of-fit tests, t-tests, ANOVA, correlation, and regression.

- R is a table-stakes language for data science and bioinformatics.
- The tidyverse makes complex data pipelines readable and maintainable.
- ggplot2 produces publication-quality figures with far less effort than most alternatives.
- R's statistical ecosystem is unmatched for the tests used in biological research

Course Learning Outcomes

By the end of this course, you will be able to:

- Compare the syntax and structure of base R with the principles of the tidyverse.
- Use dplyr to wrangle and clean messy biological datasets using a readable, pipeable grammar.
- Use ggplot2 to produce layered, publication-quality visualizations.
- Conduct and interpret standard parametric and nonparametric hypothesis tests.
- Apply ANOVA, post-hoc tests, correlation, and linear regression to biological data.
- Check statistical assumptions rigorously before choosing and running any test.

You will be comfortable using the tidyverse to wrangle/clean data, produce publication-quality ggplot2 figures, and conduct and interpret standard parametric and nonparametric statistical tests in R.

Pre-requisites:

- Completion of Introduction to R I (or demonstrated equivalent competency) is required.
- Biostatistics I (or equivalent) is strongly recommended. This course focuses on using R to apply statistical methods, not on teaching statistics from first principles. Brief refreshers are provided at the start of relevant modules, but prior familiarity is assumed.

Course Overview

Introduction to R I and Introduction to R II form a complete, two-course sequence. Together they take you from zero programming experience to conducting, visualizing, and statistically interpreting biological data analyses in a fully reproducible workflow.

Introduction to R I

"Learn R as a programming language"

- Base R syntax & data types
- Data structures: vectors, lists, data frames, matrices
- Control flow, loops, custom functions
- Data wrangling & cleaning
- Base R visualization & descriptive statistics

Introduction to R II

"Use R as a data-analysis platform"

- Tidyverse philosophy & dplyr
- ggplot2 for publication-quality visualization
- Hypothesis testing (Chi-squared, Fisher's exact, t-tests)
- ANOVA & nonparametric alternatives
- Correlation & linear regression

Undergraduate Credit Track

ABOUT

This section applies only to students enrolled for university credit through the University of Maine. Internal-only JAX participants (no credit hours) follow the standard track and do not submit the items listed here.

Credit Requirements

1 credit unit (approximately 45 hours total: 12.5 hours contact time + independent project work)

- Attend all five modules and actively participate in in-class exercises.
- Complete all five standard module assignments (same as internal students).
- Complete the five undergraduate credit questions (one per module) using the JAXMouse dataset.
- Submit each module's Project Milestone file by the date posted on Canvas.
- Your Module 5 compiled report forms the R I component of the semester project, continued in R II.

Shared Dataset: JAXMouse Gene Expression (Simulated)

All undergraduate credit questions across both courses use a single simulated RNA-seq-style dataset: JAXMouse_GeneExpression_Simulated.csv.

Column	Description
SampleID	Unique sample identifier (e.g., Brain_CTL_Rep1)
Tissue	Factor: Brain, Liver, Muscle
Treatment	Factor: Control, Treated
Replicate	Integer: 1, 2, or 3
Sex	Factor: Male, Female
Age_weeks	Numeric: mouse age at sacrifice (weeks)
Gene	Character: gene symbol (e.g., Actb, Gapdh, Brca1)
Expression	Numeric: log2 CPM expression value
QC_pass	Logical: TRUE / FALSE quality control flag
BatchID	Factor: Batch1, Batch2 (used to illustrate Simpson's Paradox)

Project Milestones: One Per Module

Module	Milestone Deliverable	Due
Module 1	JAXMouse_clean.csv (factors converted, columns verified)	*See GitHub*
Module 2	JAXMouse_gene_summary.csv (mean, SD, outlier flag per gene)	*See GitHub*
Module 3	JAXMouse_QCpass.csv (final clean, QC-filtered dataset)	*See GitHub*
Module 4	JAXMouse_baseR_plots.pdf (base R visualization suite)	*See GitHub*
Module 5	JAXMouse_RI_Analysis report (compiled Quarto HTML)	*See GitHub*

Schedule & Content

Module	Topic & Content	Resources
	MODULE 0 — Setup & Orientation *Q: Are we ready to pick up where Intro to R I left off?* *A: Verify that R/RStudio are installed and run correctly and refresh the core R I skills we will build upon throughout this course.*	
0	**Learning Objectives** • Confirm R and RStudio (Posit) are installed and functional. • Recall key R I concepts: data structures, slicing, tapply(), base R plots. • Review the course overview and confirm prerequisites are met. **In-Class Activities** Self-check: run a short R I review script provided on Canvas before Day 1.	**Installation** • JAX Software Catalogue (internal) **General R Reference** • R for Data Science (Hadley Wickham) • Intro to R (online book) **R I refresher** • R for Data Science Stats Flowchart
	MODULE 1 — Reviewing the Foundations: from R I to R II *Q: How do we leverage our foundational coding skills to efficiently interrogate a new biological dataset?* *A: We consolidate R I competencies, like data import, slicing, tapply(), base R plots, before layering the tidyverse on top. Everything you learned in R I is still valid; the tidyverse simply extends it.*	
1AB	**Learning Objectives** • Reload and inspect datasets using the full suite of base R tools (summary(), str(), head(), dim()). • Apply slicing with conditionals, subset(), and the *apply family. • Produce scatterplots, histograms, and mosaic plots in base R. • Articulate what you find clunky or repetitive about base R syntax, this motivates the tidyverse. • Understand the R II workflow: **Tidy → Transform → Visualize → Model → Interpret** **In-Class Activities** • Produce graphs with manipulated datasets in each of the above categories. • Side-by-side comparison: base R approach vs. a preview of the tidyverse approach.	**Resources:** • **ggplot2 cheat sheet** Data Visualization with ggplot2 **Assignment for Certificate of Completion:** See GitHub Course Site.
	MODULE 2 — Introduction to the Tidyverse: Tidy Data & dplyr *Q: What exactly makes data "tidy," and why does it transform the way we write analytical code?* *A: We transition from the clunkiness of base R to the elegance of the tidyverse's grammar of data manipulation, using the dplyr package.*	
2AB	**Learning Objectives** • Define "tidy data" (one observation per row, one variable per column) and identify violations in real datasets. • Work with tibbles and understand how they differ from base R data frames.	**Tidyverse guides** • Tidy Data Tutor (visual) • R4DS — Hadley Wickham • GenZ dplyr explainer

	• Use the core dplyr verbs: filter(), select(), mutate(), arrange(), group_by(), summarize(). • Chain operations with the pipe operator (|> or %>%) for readable, sequential pipelines. • Pass a dplyr result directly into ggplot2 using the pipe. **In-Class Activities** • Manipulate and graph a built-in dataset using a dplyr + ggplot2 pipeline. • Clean a messy dataset using tidyverse tools.	**Assignment for Certificate of Completion:** See GitHub Course Site.
MODULE 3 — The Grammar of Visualization: ggplot2 *Q: How do we move beyond charts to construct layered, publication-ready visualizations that expose the truth in our data?* *A: We compare visualizations made in base R to the far more flexible and attractive results from ggplot2, understanding the "grammar of graphics" that makes ggplot2 composable.*		
3AB	**Learning Objectives** • Explain the three essential components of any ggplot2 plot: data, aesthetics (aes()), and geoms. • Build plots layer by layer: geom_point(), geom_histogram(), geom_boxplot(), geom_bar(). • Modify aesthetics for color, fill, size, shape, and alpha. • Add facets (facet_wrap(), facet_grid()) to show data across subgroups. • Apply themes and custom labels for publication-ready output. • Compare the same visualization produced in base R vs. ggplot2 and critique both. **In-Class Activities** • Recreate base R Module 4 visualizations from R I using ggplot2 for a direct comparison. • Explore a new dataset and produce a ggplot2 visualization suite.	**ggplot2 cheat sheet** • Data Visualization with ggplot2 **Further reading** • R4DS — Data visualisation **Assignment for Certificate of Completion:** See GitHub Course Site.
MODULE 4 — Hypothesis Testing: Binomial, Chi-Squared, t-Tests & Survival Analysis *Q: How do we translate a biological question into a mathematically rigorous, code-based hypothesis test?* *A: We establish a universal four-step framework for hypothesis testing and apply it to an escalating series of tests: from binomial proportions through goodness-of-fit, t-tests, and Kaplan-Meier survival curves.*		
4AB	**Learning Objectives** • Apply the four-step hypothesis testing framework: state H_0/H_a → choose and run the test → compare p-value to α → conclude. • Distinguish Type I and Type II errors and understand statistical power. • Build a binomial distribution and apply binom.test() and prop.test(). • Apply chi-squared and Fisher's exact tests to contingency tables. • Test normality using QQ plots and the Shapiro-Wilk test. • Conduct one-sample, two-sample, and paired t-tests; identify the correct test for the experimental design.	**Reference** • Hypothesis testing (Wikipedia) • Type I & II errors (visual) • Power demo 1 • Power demo 2

	• Apply nonparametric alternatives (Wilcoxon test, Welch's t-test) when normality assumptions are violated. • Construct and interpret Kaplan-Meier survival curves. • Calculate and interpret ROC/AUC for diagnostic classifiers. **In-Class Activities** • Formally test a hypothesis on the Bumpus sparrow dataset (binomial test). • Reproduce a ROC/AUC example using the pROC library. • Produce a Kaplan-Meier curve for the built-in leukemia dataset.	**Assignment for Certificate of Completion:** See GitHub Course Site.
MODULE 5 — ANOVA, Correlation & Linear Regression *Q: How can we reliably predict outcomes and map complex relationships between multiple variables while accounting for biological noise and outliers?* *A: We extend hypothesis testing from two-group comparisons to multi-group ANOVA and continuous relationships via correlation and linear regression, always checking assumptions and investigating residuals.*		
5AB	**Learning Objectives** • Test homogeneity of variances with bartlett.test() before running ANOVA. • Run a one-way ANOVA and apply TukeyHSD() for post-hoc pairwise comparisons. • Apply the nonparametric alternative to ANOVA: kruskal.test(). • Distinguish between correlation (measuring association) and regression (predicting values). • Calculate Pearson's r and its nonparametric alternatives (Spearman, Kendall). • Fit a simple linear regression, interpret coefficients, and calculate confidence intervals. • Produce and interpret a residual plot to identify outliers and check model assumptions. **In-Class Activities** • Cuckoo egg ANOVA: test whether egg size differs across host species. • Mammal brain-body size regression: log-transform, fit, interpret residuals, identify outliers.	**Simulations** • Visual ANOVA (Wolfram) • ANOVA video • Regression fit demo • Correlation vs. Regression game **Assignment for Certificate of Completion:** See GitHub Course Site.

Further Learning & Enrichment Resources

The resources below supplement the course content. None are required; they are provided for students who want to go deeper.

Resource	URL / Notes
R for Data Science (R4DS)	https://r4ds.hadley.nz/

tidyverse documentation	https://www.tidyverse.org/
ggplot2 cheat sheet	https://posit.co/wp-content/uploads/2022/10/data-visualization-1.pdf
Tidy Data Tutor (visual dplyr)	https://tidydatatutor.com
Statistics flowchart	https://www.statsflowchart.co.uk
Visual ANOVA (Wolfram)	https://demonstrations.wolfram.com/VisualANOVA
Correlation vs. Regression game	http://shiny.stat.calpoly.edu/Corr_Reg_Game/